



Koneru Lakshmaiah Education Foundation

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CAR RENTAL & MOBILITY PLATFORM

Simple Explainer

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Agenda

- What Is This Project?
- Why Different Databases?
- How a Request Flows
- The One “Smart” Part
- Summary

What Is This Project?

A backend system for a car rental app — think Zoomcar or Revv — that handles bookings, vehicle listings, live availability, and recommendations.

Why It's a Good Database Project

Instead of using one database for everything, we deliberately use a different database for each kind of data — because each type of data behaves differently.

Bookings, payments, invoices

PostgreSQL

Needs to be 100% correct — no double bookings, no lost payments

Vehicle listings, photos, reviews

MongoDB

Messy and varied — a bike listing differs from an SUV listing

Recommendations

pgvector

Finds “similar” vehicles — same idea as Spotify/Netflix

How a Request Flows

1

User opens the app

Hits FastAPI, which checks who they are (login/roles) and routes the request.

2

Needs a live update?

“Is this car still available right now” goes to a Node.js service using Socket.io/Redis for real-time push.

3

Inventory, pricing, or placing an order?

Goes to a Spring Boot service.

4

Either way

Both services read and write to the databases from the previous slide.

The One “Smart” Part

Booking a car touches three things at once: check inventory, hold the car, charge payment.

If payment fails halfway, you can't leave the car “held” forever.

The Saga Pattern

A way to undo the earlier steps if a later step fails, without locking the whole database.

Production-Ready, Too

Everything runs in Docker containers, deployed via Kubernetes, and monitored using Prometheus/Grafana. GitHub Actions auto-tests and deploys on every code change.

Abstract

- The Car Rental and Mobility Platform is a modern digital system designed to simplify vehicle rental and mobility services by connecting customers, vehicles, bookings, payments, and service providers through a unified platform.
- The system provides efficient management of vehicle information, specifications, images, reviews, pricing, and rental availability.
- The system addresses the need for transactional consistency (bookings & payments), flexible catalog data (listings & reviews), real-time tracking, and similarity-based recommendations at scale.
- The platform provides real-time vehicle availability and tracking updates, helping customers and service providers manage vehicles more efficiently.
- It provides role-based access for customers, rental agents, and administrators through a secure API gateway.
- PostgreSQL manages structured transactional data, MongoDB handles flexible vehicle catalog & review records, and pgvector powers similarity search for recommendations.
- The backend is designed for containerized, service-oriented deployment with full observability, applying the database and distributed backend concepts covered in the course.